Class 12 Practice Sheet

Nov 4, 2026 · @Laura

Directions

These are the same 30 exercises as the Class 12 app, in the same order, so you can work on paper and follow along in class. The answer key is at the end.

Name: _____ Date: _____

Part 1: Names, predicates, and quantity words

Name the expression

For each sentence, say what kind of expression the **bold** phrase is: proper name, definite description, pronoun, quantity expression, or none of these.

1. Ana's only sister is a pilot.
Kind of expression: _____

2. Nobody finished the puzzle.
Kind of expression: _____

3. How many places does the predicate 'x sits between y and z' have?
Number of places: _____

Quick check

Circle the best answer.

4. Which of these is a singular term?

A. 'the tallest mountain in Peru'
B. 'nobody'
C. 'is taller than'
D. 'everyone'

5. Start with 'Lisbon is west of Madrid'. Which can you NOT get by deleting singular terms?

A. ‘x is west of y’
B. ‘Lisbon is x of Madrid’
C. ‘x is west of Madrid’
D. ‘Lisbon is west of x’

6. ‘Every member of the chess club plays online. Mia is a member of the chess club. So Mia plays online.’ Why does sentential logic call this valid argument invalid?

A. Each sentence becomes its own unrelated letter, so the shared name and predicates disappear
B. Sentential logic can’t handle three sentences
C. Because it really is invalid
D. Sentential logic only works when the premises are true

Part 2: Reading formulas

Read a formula

For each formula: (a) Is it atomic, truth-functional, or quantified? (b) What is its main operator? (c) Which variables are free? (d) Is it a sentence?

7. (∃y)Fy & Gx
(a) _____ (b) _____ (c) _____ (d) yes / no

8. ~(∀x)(Fx ⊃ (∃y)Rxy)
(a) _____ (b) _____ (c) _____ (d) yes / no

Substitution instances

Circle every substitution instance of the sentence. If it has none, write “none”.

9. (∃x)(Hx ⊃ (∃z)Hz)

a) Hc ⊃ (∃z)Hz
b) Ha ⊃ (∃z)Hz
c) (∃x)(Hc ⊃ (∃z)Hz)

10. (∃y)(∃x)(Hx ⊃ Hy)

a) (∃y)(∃x)(Hx ⊃ Ha)
b) Hb ⊃ Ha
c) (∃x)(Hx ⊃ Hb)
d) (∃x)(Hx ⊃ Ha)

Quick check

Circle the best answer.

11. Which symbol is an individual constant?

A. x
B. m
C. w
D. M

12. Which is NOT a formula?

A. Lbz
B. B
C. (∃b)Hb
D. (∃x)Hx

13. Which of these is NOT a formula?

A. Lxy
B. (∃y)Gx
C. (∃x)(Fx & Gy)
D. ~(∀z)Hz

Part 3: All, no, and some

Symbolize a sentence

For each sentence: (a) name its form (A, E, I, O, or all and only), then (b) symbolize it with the key given.

14. No doctors are runners.
UD: people. Dx: x is a doctor. Rx: x is a runner.
(a) Form: _____ (b) _____

15. At least one poet is a teacher.
UD: people. Px: x is a poet. Tx: x is a teacher.
(a) Form: _____ (b) _____

16. Not all students are athletes.
UD: people. Sx: x is a student. Ax: x is an athlete.
(a) Form: _____ (b) _____

17. Only plants are ferns.
UD: living things. Px: x is a plant. Fx: x is a fern.
(a) Form: _____ (b) _____

Quick check

Circle the best answer.

18. Which sentence is the contradictory of ‘Some students passed’?

A. ‘Some students did not pass.’
B. ‘No students passed.’
C. ‘All students passed.’
D. ‘Only students passed.’

19. UD: the students in PHIL 270. Px: x passed. Symbolize ‘Everyone passed.’

A. $(\forall x)(Sx \supset Px)$
B. $(\exists x)Px$
C. $(\forall x)Px$
D. Pe

20. Every student passed. Is ‘Some students passed’ true?

A. Yes. ‘Some’ means at least one
B. It can’t be decided
C. No. ‘Some’ means more than one
D. No. ‘Some’ implies ‘not all’

Part 4: Fine-tuned symbolization

Quantifier order

Symbolize each sentence. UD: people.

21. Everyone likes someone.
Key: Lxy: x likes y.
Symbolization: _____

22. Someone trusts everyone.
Key: Txy: x trusts y.
Symbolization: _____

23. There is someone whom everyone admires.
Key: Axy: x admires y.
Symbolization: _____

24. No one knows everyone.
Key: Kxy: x knows y.
Symbolization: _____

Move a quantifier

Rewrite each sentence so one quantifier covers the whole sentence. Which quantifier does the equivalent sentence start with: (∀x), (∃x), or can it not be moved?

25. Fb ⊃ (∀x)Hx
Starts with: _____ Rewritten: _____

26. (∃x)Gx ⊃ Hc
Starts with: _____ Rewritten: _____

27. (∃x)Hx ≡ Fb
Starts with: _____ Rewritten: _____

Quick check

Circle the best answer.

28. UD: people and songs. Px: x is a person. Kxy: x knows y. Symbolize ‘Someone knows both the school song (s) and the anthem (n).’

A. (∃x)(Kxs & Kxn)
B. (∃x)[Px & (Kxs & Kxn)]
C. (∃x)(Px & Kxs) & (∃x)(Px & Kxn)
D. (∀x)[Px ⊃ (Kxs & Kxn)]

29. Tomas hopes to find a four-leaf clover. Any one would do, and there may be none in the field. Which is best?

A. (∃y)(Cy & Hty)
B. Htc, with c: the clover
C. (∀y)(Cy ⊃ Hty)
D. Ft, with Fx: x hopes to find a four-leaf clover

30. UD: people. Lxy: x likes y. Which says ‘There is someone whom everyone likes’?

A. (∃x)(∀y)Lyx
B. (∀x)(∃y)Lxy
C. (∃x)(∀y)Lxy
D. (∀y)(∃x)Lyx

Answer key

Part 1

1. Definite description. ‘Only’ limits it to at most one person, so it is a definite description.

2. Quantity expression. It sits where a name could go, but it names no one. It says there is no person who finished.

3. 3 places. The gaps are x, y, and z.

4. A. A singular term picks out one thing. ‘Everyone’ and ‘nobody’ are quantity words; ‘is taller than’ is a predicate.

5. B. Only singular terms can become gaps. ‘West’ is part of the predicate.

6. A. In sentential logic it is just C, M, so P. The argument is valid because of the shared name ‘Mia’ and the shared predicates, which only predicate logic can show.

Part 2

7. (a) Truth-functional. (b) &. (c) x. (d) No, because x is free.

8. (a) Truth-functional. (b) ~. (c) None. (d) Yes, it has no free variables.

9. a and b. Not c: the initial quantifier has to be dropped, and keeping it leaves a quantifier with no x to bind.

10. c and d. Not a: the initial quantifier has to be dropped. Not b: it drops two quantifiers, and only the initial one goes.

11. B. Constants are lowercase a through v. Lowercase w, x, y, z are variables, and capitals are predicates.

12. C. Quantifiers take variables, not constants. The capital letter B on its own is a sentence letter, which is an atomic formula.

13. B. A quantifier has to bind something: (∃y) can only go in front of a formula that contains y. Gx has no y.

Part 3

14. (a) E (no). (b) (∀x)(Dx ⊃ ~Rx), or equivalently ~(∃x)(Dx & Rx)

15. (a) I (some). (b) (∃x)(Px & Tx)

16. (a) O (some … not). (b) (∃x)(Sx & ~Ax), or equivalently ~(∀x)(Sx ⊃ Ax)

17. (a) A (all). (b) (∀x)(Fx ⊃ Px). ‘Only plants are ferns’ means ‘All ferns are plants’, so the term after ‘only’ goes in the consequent.

18. B. Contradictories always have opposite truth values. E and I are contradictories, and so are A and O.

19. C. When the UD is only students, no ‘student’ predicate is needed.

20. A. In logic, ‘some’ means at least one and does not imply ‘not all’.

Part 4

21. (∀x)(∃y)Lxy. Each person x likes some y or other. Different people may pick different ys, so (∀x) comes first.

22. (∃x)(∀y)Txy. One person x trusts every y, so (∃x) comes first and x is in the first place.

23. (∃x)(∀y)Ayx. One person x is such that every y admires x, so (∃x) comes first and x goes in the second place.

24. ~(∃x)(∀y)Kxy. It is not the case that someone knows everyone. (∀x)(∃y)~Kxy is an equivalent way to write it.

25. (∀x). Rewritten: (∀x)(Fb ⊃ Hx). The quantifier is in the consequent, so it keeps its type.

26. (∀x). Rewritten: (∀x)(Gx ⊃ Hc). The quantifier is in the antecedent, so it flips: ∃ becomes ∀. Compare ‘If anyone sings, Ana sings’: it means ‘Everyone is such that if they sing, Ana sings.’

27. It can’t be moved. A quantifier on one side of ≡ can’t, in general, be broadened over the whole biconditional. Neither (∀x) nor (∃x) gives an equivalent sentence.

28. B. One person knows both. C allows two different people, and A lets ‘someone’ pick out a song.

29. D. No particular clover is sought, and there may be none. Put that language inside a one-place predicate.

30. A. One person x is such that every y likes x, so (∃x) comes first and x goes in the second place.